

Let u and v be primitive elements such that $t_1^n h_1 t_1^n h_1^{-1} \in \langle u \rangle$ and $t_2^m h_2 t_2^m h_2^{-1} \in \langle v \rangle$. Applying Proposition 5.1 to y and using Lemma 2.2, we obtain that

$$\begin{aligned} \mathbb{E}_u \otimes \mathbb{E}_v(y) &= c_{e,e}^2 e \otimes e + c_{t_1^n, e}^2 t_1^n h_1 t_1^n h_1^{-1} \otimes e + c_{e, t_2^m}^2 e \otimes t_2^m h_2 t_2^m h_2^{-1} + c_{(t_1^n, t_2^m)}^2 t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1} \\ &= 0 + 0 + 0 + c_{(t_1^n, t_2^m)}^2 t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1} \in \mathcal{A}. \end{aligned}$$

Since $c_{(t_1^n, t_2^m)} \neq 0$, we get that $t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1} \in \mathcal{A}$. Now, replacing h_1, h_2 by h_1^2 and h_2^2 respectively, and using the same argument (noting that $\langle t_i, h_i^2 \rangle \cong \mathbb{F}_2$ for each $i = 1$ and 2), we get that

$$t_1^n h_1^2 t_1^n h_1^{-2} \otimes t_2^m h_2^2 t_2^m h_2^{-2} \in \mathcal{A}.$$

Consequently,

$$\begin{aligned} (h_1 \otimes h_2)(h_1 t_1^{-n} h_1^{-1} t_1^n \otimes h_2 t_2^{-m} h_2^{-1} t_2^m)(h_1^{-1} \otimes h_2^{-1}) \\ = h_1^2 t_1^{-n} h_1^{-1} t_1^n h_1^{-1} \otimes h_2^2 t_2^{-m} h_2^{-1} t_2^m h_2^{-1} \\ = h_1^2 t_1^{-n} h_1^{-2} h_1 t_1^n h_1^{-1} \otimes h_2^2 t_2^{-m} h_2^{-2} h_2 t_2^m h_2^{-1} \\ = (t_1^n h_1^2 t_1^n h_1^{-2} \otimes t_2^m h_2^2 t_2^m h_2^{-2})^{-1} (t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1}) \in \mathcal{A}. \end{aligned}$$

Therefore, $h_1 t_1^{-n} h_1^{-1} t_1^n \otimes h_2 t_2^{-m} h_2^{-1} t_2^m \in \mathcal{A}$. Hence,

$$t_1^{2n} \otimes t_2^{2m} = (t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1})(h_1 t_1^{-n} h_1^{-1} t_1^n \otimes h_2 t_2^{-m} h_2^{-1} t_2^m) \in \mathcal{A}.$$

Therefore, $\mathbb{E}_{t_1} \otimes \mathbb{E}_{t_2}(a)(h_1 t_1^{2n} h_1^{-1} \otimes h_2 t_2^{2m} h_2^{-1}) \in \mathcal{A}$. Observe that

$$\mathbb{E}_{t_1} \otimes \mathbb{E}_{t_2}(a)(h_1 t_1^{2n} h_1^{-1} \otimes h_2 t_2^{2m} h_2^{-1}) = \sum_{k, l \in \mathbb{Z}} c_{t_1^k, t_2^l} t_1^k h_1 t_1^{2n} h_1^{-1} \otimes t_2^l h_2 t_2^{2m} h_2^{-1}.$$

Let u' and v' be primitive elements such that $t_1^n h_1 t_1^{2n} h_1^{-1} \in \langle u' \rangle$ and $t_2^m h_2 t_2^{2m} h_2^{-1} \in \langle v' \rangle$. Using Proposition 5.1, and Lemma 2.2, we see that

$$t_1^n h_1 t_1^{2n} h_1^{-1} \otimes t_2^m h_2 t_2^{2m} h_2^{-1} = \frac{1}{c_{t_1^n, t_2^m}^2} \mathbb{E}_{u'} \otimes \mathbb{E}_{v'} (\mathbb{E}_{t_1} \otimes \mathbb{E}_{t_2}(a)(h_1 t_1^{2n} h_1^{-1} \otimes h_2 t_2^{2m} h_2^{-1})) \in \mathcal{A}.$$

We recall that $t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1} \in \mathcal{A}$. Therefore,

$$\begin{aligned} (t_1^n h_1 t_1^{2n} h_1^{-1} \otimes t_2^m h_2 t_2^{2m} h_2^{-1})(t_1^n h_1 t_1^n h_1^{-1} \otimes t_2^m h_2 t_2^m h_2^{-1})^{-1} \\ = t_1^n h_1 t_1^{2n} h_1^{-1} h_1 t_1^{-n} h_1^{-1} t_1^{-n} \otimes t_2^m h_2 t_2^{2m} h_2^{-1} h_2 t_2^{-m} h_2^{-1} t_2^{-m} \\ = t_1^n h_1 t_1^n h_1^{-1} t_1^{-n} \otimes t_2^m h_2 t_2^m h_2^{-1} t_2^{-m} \\ = (t_1^n h_1 \otimes t_2^m h_2)(t_1^n \otimes t_2^m)(t_1^n h_1 \otimes t_2^m h_2)^{-1} \in \mathcal{A}. \end{aligned}$$

Since $\mathcal{A}$ is invariant, it follows that $(g, h) = t_1^n \otimes t_2^m \in \mathcal{A}$. The claim follows. Since property-AP is preserved under extensions, Γ has property-AP, and we can argue as before to conclude the proof. $\square$

Remark 5.5. Our proof will show that $\Gamma = \Gamma_1 \times \Gamma_2 \times \dots \times \Gamma_n$ has C^* -ISR property if every Γ_i is a torsion-free non-amenable acylindrically hyperbolic group with property-AP.